

Synaptic Plasticity as Short-Term Memory

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Core idea. Synaptic plasticity is the ability of a connection between neural units to change its effective strength as activity occurs. In a computational short-term-memory view, these rapidly changing synaptic states can encode recent associations without requiring every past input to remain explicitly stored as a separate item. The resulting memory is useful precisely because it is fast and temporary: the same mechanism that permits rapid adaptation also creates vulnerability to decay and interference.

Why this matters now. Modern sequence models need ways to adapt to recent context without treating every interaction as permanent parameter learning. Recent machine-learning research has revisited plasticity and fast synaptic state as a computational substrate for short-term memory. Rodriguez, Guo and Moraitis (ICML 2022) introduced Short-Term Plasticity Neurons in which synapses carry a transient state through time and showed strong performance on associative retrieval and other dynamic tasks. This places plasticity in a current ML research context rather than treating it only as a neuroscience metaphor.

Mechanism demonstrated by this submission. Our interactive artifact uses a deliberately small associative matrix. A cue activates a pre-synaptic unit and a target activates a post-synaptic unit. A simplified Hebbian write increases the corresponding synaptic strength, while a retention factor decays existing fast memory. In the toy model, the update is $w(t+1) = r \cdot w(t) + \eta \cdot \text{pre}(t) \cdot \text{post}(t)$. A learner can change the learning rate and retention, write associations, advance time, and query the memory. A conflicting write can become the strongest trace and cause an incorrect recall. The expected association is shown beside the model output, so the learner can test the claim rather than merely read it.

BDH connection. The Dragon Hatchling (BDH) is a brain-inspired post-Transformer architecture. Its primary paper states that BDH working memory during inference relies on synaptic plasticity with Hebbian learning using spiking neurons, and reports strengthening of individual synapses when the model processes particular concepts. This makes synaptic plasticity a native architectural mechanism in BDH, not a decorative analogy. Our artifact is intentionally **not** presented as BDH: it is an independent toy model that isolates the memory mechanism so its variables are visible and manipulable. The BDH module should be read alongside the primary paper's architecture and equations.

Comparison and trade-offs. Conventional parameter learning changes slow model parameters and is normally expensive to perform at inference time. A fast synaptic state can instead adapt quickly to recent activity, but its capacity is limited and new writes can interfere with earlier traces. Short-term plasticity therefore creates a stability–plasticity trade-off: higher write strength improves rapid adaptation but can make recent events dominate; stronger retention preserves older traces but can reduce flexibility. The artifact makes this trade-off observable in seconds.

Evidence and limits. The published evidence supports the existence and usefulness of computational plasticity mechanisms; it does not prove that this four-by-four toy system reproduces BDH behavior or that the toy rule is biologically complete. The most important limitation is therefore evidence level: the demo is an educational computational model, while BDH claims are taken from the published primary architecture paper. A further open question is how much useful information can be stored in rapidly changing synapses before interference overwhelms retrieval.

Primary sources: Kosowski et al. (2025), *The Dragon Hatchling*; Rodriguez et al. (2022), *Short-Term Plasticity Neurons Learning to Learn and Forget*; Brito & Gerstner (2024), *Learning what matters*. Full links are in REFERENCES.md.